

1 Introduction

Main points:

- Define SPE's
- Define Coexistence
- Growing interest in the literature
- tie to climate change
- basic sentence about how it works? (ie weaker competitors go earlier)
- BUT. it doesn't work because **phenology** is not a trait
- in this paper we....say what we will do. One goal Goal demonstrate a simple way to better incorporate realistic physiology into models of coexistence and to integrate experiments and models

2 How do phenological differences arise?

- Discuss cues
- define phenological sensitive
- state that interaction between sensitivity and environment result in **realize phenological difference**
- Explain figure 1 and use as example

3 How we (currently) model phenology is coexistence (and why it doesn't work)

- a paragraph discussing how it's been done
- a paragraph about non-stationarity and shifting environmental means and why the current approaches break down under the conditions

4 A new hope

not sure if this should be its own section or transition from the previous one

- we need a model to do this
- here we show one of the simplest ways to do it by adding an intermediate between chillign nad response
- maybe a brief description of the approach with refrence to supporting information for details
- even with such a simple approach we see really interesting things emerge about both the relationships between env., phenological sensitivities, and realized phenology as well as between phenological sensitivity and coexistence

Snappy title about results and inference

4.1 A shifting environment shifts the relationship between phenological sensitivity difference and realized phenology differences

- Basically discuss a 3 panel figure. (histograms of chilling distributions and current figures 2c and d)

4.2 A shifting environment shifts the tradeoff between phenological sensitivity and competition

- The trade off between realized phenology and R_{star} is the same
- Shift in the tradeoff between phenosensitivity and R_{star}
- explicitly highlight that this means different pairs of species coexist

5 A path forward

Our model is only the beginning, and while it is a noteworthy advance worthy of being published in a nature subsidiary or equivalently high impact journal, there is still a long way to go to forecast these dynamics. Here are some critical aspects. **I think I need help here. All of my thoughts are pretty germination/chilling specific. I could use some idea for contextualizing this more broadly in phenology, and coexistence**

1. multiple interacting cues
2. add risk of being early
3. trait and density dependent mechanisms (rudoff paper)/ ie, we vary germination fraction and phenology.
4. other life histories, i.e, perennials, trees.

Figures

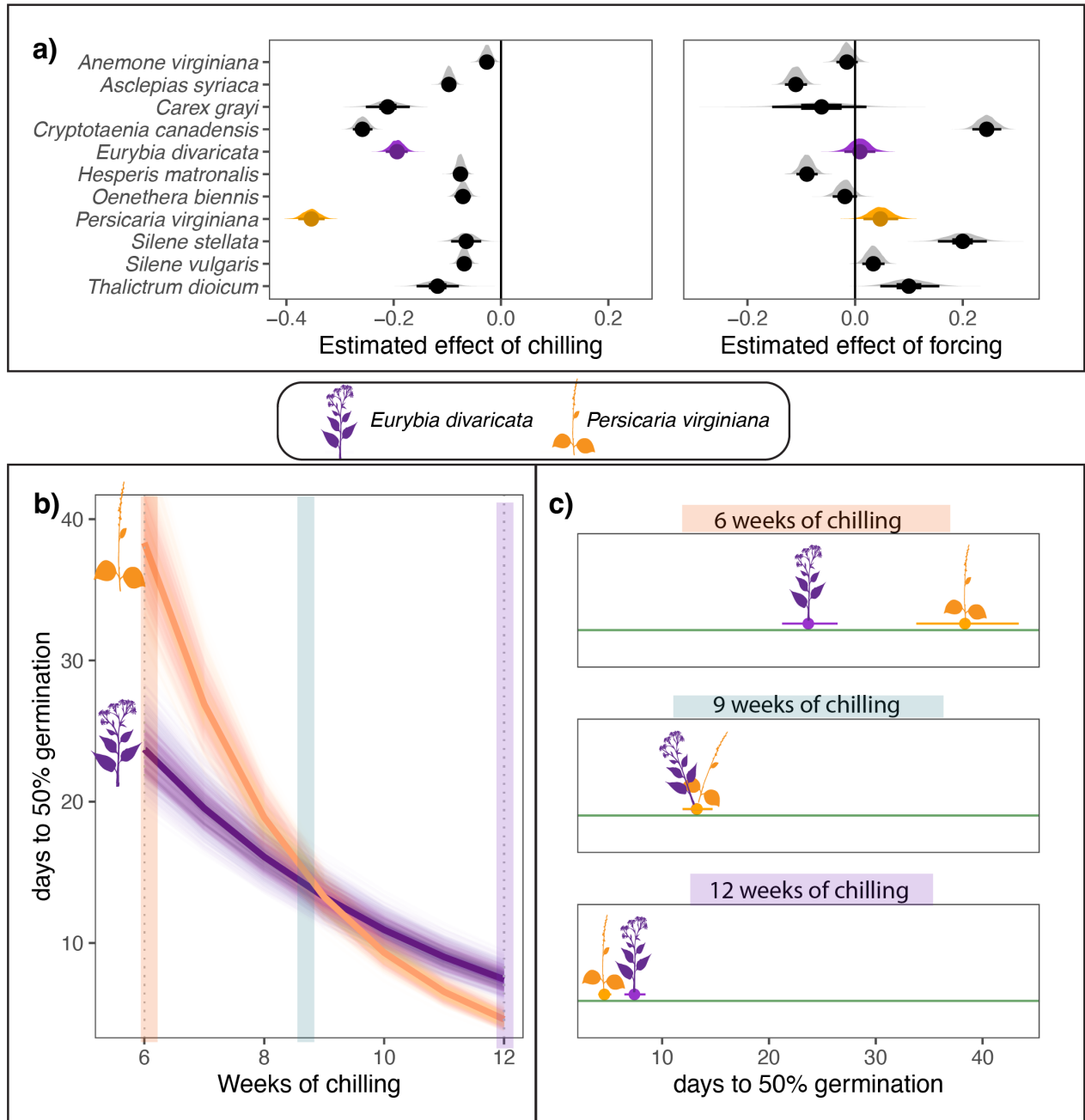
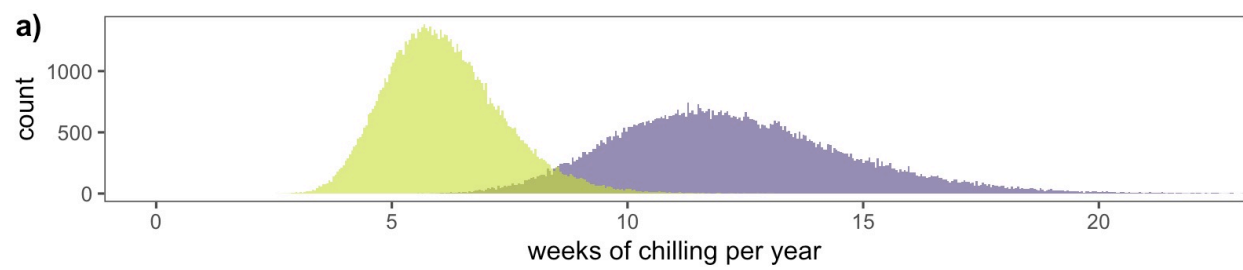


Figure 1: **Co-occur species can differ substantially in their phenological sensitivity (Δ phenological event/ Δ climate) to the environment.** Panel a) estimates the phenological sensitivity of germination timing to chilling and forcing for eleven herbaceous species of eastern North America. Points represent the mean estimate and thick and thin bars represent the 50 and 90% uncertainty intervals with the full posterior distribution displayed. In b) and c) we highlight the phenological sensitivity to chilling of two species *Eurybia divaricata* and *Persicaria virginiana* that are commonly observed in competition. b) depicts the full sensitivity of each species to chilling of each species, predicted from Bayesian accelerated failure time models, with thick lines representing the mean estimate and lighter lines depict 1000 draws from the posterior distribution to represent uncertainty. Panel c) illustrates how the species' different sensitivities depicted in b) manifest in contrasting realized germination differences under different climate conditions. Point represent the predicted mean number of days to 50% germination and bars the 90% uncertainty intervals. Under six weeks of chilling *E. divaricata* germinates well before *P. virginiana*. Under nine weeks of chilling the species germination at roughly the same time, and under twelve weeks of chilling *P. virginiana* is the first to germinate.



average chilling period 12 weeks 6 weeks

